CodeAlpha Artificial Intelligence Internship

Project Documentation and Submission Pack

Student: Samkelisiwe Beverly Khuzwayo

Student ID: CA/DF1/315315

1. Project Overview

This document describes two beginner-level Artificial Intelligence projects prepared in line with the supplied CodeAlpha task brief: a Language Translation Tool and an FAQ Chatbot.

2. Task 1 — Language Translation Tool

Objective: Build a user interface that accepts text, lets the user choose source and target languages, sends the text to a translation service, and displays the translated result.

Implementation: Python and Streamlit provide the interface. The deep-translator package connects to Google Translate. The app includes source and target language selectors, text input, result display and basic error handling.

How to run: Open the Task_1_Language_Translation_Tool folder, install packages using pip install -r requirements.txt, then run streamlit run app.py.

3. Task 2 — Chatbot for FAQs

Objective: Collect FAQs, process text, compare a user's question with the FAQ questions, and display the most relevant answer.

Implementation: The project uses a sample student-support FAQ dataset. scikit-learn's TF-IDF vectorizer converts questions into numerical features, and cosine similarity identifies the closest match. Streamlit provides the chat interface.

How to run: Open the Task_2_FAQ_Chatbot folder, install packages using pip install -r requirements.txt, then run streamlit run app.py.

4. Testing Plan

For the translation tool, test normal text, automatic source detection, same-language selection, empty input and internet failure. For the FAQ chatbot, test an exact FAQ, a reworded FAQ, an unrelated question and empty input. Record the actual outcomes after running the applications.

5. Limitations

The translation tool relies on an external online translation service, so internet access and service availability are required. The FAQ chatbot only answers questions represented in its sample FAQ dataset and is not a general-purpose conversational AI system.

6. Submission Notes

Upload the two complete task folders, this documentation, and genuine screenshots or a demonstration video to Google Drive. Check that the shared link can be opened by the intended reviewer. The supplied CodeAlpha brief also requires use of the official submission form shared in the WhatsApp group. LinkedIn posting and form submission must be completed by the student.

Prepared project materials. Run and verify both applications before final submission.